

The Pedigree Diagram Tab

Overview

The Pedigree Browser’s **Diagram** tab renders a pedigree as an interactive family-tree diagram, following the same mating-unit convention traditional pedigree charts (and the `kinship2` R package) use: a mate’s own matings render as a small connector between the two parents, with a line down to their shared children, rather than two independent lines running straight from each parent. This article walks through everything the tab shows and every control above it, using the Shiny app (`runGeneKeepR()`) directly.

Two related resources cover the same feature from other angles:

- The **Colony Manager’s Guide** article’s “Diagram view” section places this tab in the context of the full Pedigree Browser – Table view, focal-animal trimming, and export – for a reader working through the app tab by tab.
- The **Interactive Use of `nprcgenkeepr`** vignette’s “Pedigree Diagram” section shows the same rendering built entirely from R, script by script, via the two exported functions this tab is built on (`makePedigreeMatingLayout()` and `visNetwork::visNetwork()`) – useful outside the Shiny app, e.g. for a reproducible report or a custom app.

Node shapes and the legend

Every animal is one node, shaped by sex: dot = Female, square = Male, star = Hermaphrodite, triangle = Unknown, diamond = Other/Unrecorded. A legend to the right of the diagram shows the same mapping, so it never has to be memorized.

Pedigree Browser

The processed pedigree file shown here will minimally include:

- Ego ID, Sire ID, Dam ID, Sex, Generation, and Population columns
- Other potential columns that may appear depending on information provided in the input file include: Birth Date, Exit Date, Age, Group, and Import columns may be present depending on the input file.

Focal Animals

IDs of selected focal animals may be manually entered here if analysis of all individuals is not needed. IDs may be pasted from Excel or you can browse for and select a file containing the list of focal animals.

8LKBV9, FJIB3R, GA204Z

Choose CSV file with focal animals

Browse... No file selected

Update Focal Animals

☐ Clear Focal Animals

The search field below will search all columns for matches to any text or number entered.

Display Options

☒ Display Unknown IDs

Unknown IDs, by default beginning with a capital U (the format is configurable via `setAutoldFormat()`), are created by the application for all animals with only one parent.

☒ Trim pedigree based on focal animals

Trim the pedigree to include only the ancestors and descendants of the focal animals provided.

Export Pedigree (CSV)

A population must be defined before proceeding to the Genetic Value Analysis.

Twin/Zygosity Relations (optional)

Upload a twin zygosity sidcar file (CSV or Excel) with columns id1, id2, code (MZ twin, DZ twin, or UZ twin) to mark twin pairs with a connector on the Diagram tab. Leave empty

Browse...

QC passed! 375 records processed.

No file selected

Updated focal animals: 3 IDs

Table
Diagram

Diagram Edge Style

☐ Direct ☒ Rectilinear (kinship2-style)

☐ Show Names on Diagram

☐ Show Twin Connectors

Select by id

Sex

Female

Male

Hermaphrodite

Unknown

Other / Unrecorded

Affected MZ

DZ

Export Diagram (PNG)

Figure 1: The Diagram tab on a small, real 6-animal pedigree, with its shape-to-sex legend. The two connector lines joining the top union render thicker and in a distinct vermillion color – see “Consanguineous mating marker” below.

Diagrams render up to **400 animals** under the default Rectilinear edge style (**750** if switched to Direct, below) – for a larger population, narrow the focal-animal selection first (the **Focal Animals** panel to the left of the diagram; see the Colony Manager’s Guide article for the full trimming workflow). Rectilinear’s lower cap reflects it rendering more total diagram nodes per animal (invisible routing waypoints, described below) for the same visual complexity.

An animal that mates more than once, or whose lineage loops back on itself (e.g. a consanguineous mating), appears once per mating; each occurrence after the first is joined back to its main occurrence by a curved, dashed line – visible near the top of the screenshot above, where 8LKBV9 appears twice. Hovering, clicking, or searching any occurrence behaves identically to the animal’s main occurrence (“Interacting with the diagram” below).

Diagram Edge Style: Direct vs. Rectilinear

A **Diagram Edge Style** toggle above the diagram switches the parent-to- mating-to-child connector between two routings: **Rectilinear (kinship2-style)** (the default – strict horizontal/vertical right angles throughout: a horizontal segment between the two parents, a vertical drop to the mating dot, a horizontal bar across the children, and a vertical drop into each one) and **Direct** – a single straight or lightly sloped line all the way from parent to mating dot to child.

Pedigree Browser

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Focal Animals

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8LKBV9, FJIB3R, GA204Z

Choose CSV file with focal animals

No file selected

 Update Focal Animals☐ Clear Focal Animals

The search field below will search all columns for matches to any text or number entered.

Display Options

☒ Display Unknown IDs

Unknown IDs, by default beginning with a capital U (the format is configurable via `setAutoIdFormat()`), are created by the application for all animals with only one parent.

☒ Trim pedigree based on focal animals

Trim the pedigree to include only the ancestors and descendants of the focal animals provided.

 Export Pedigree (CSV)

A population must be defined before proceeding to the Genetic Value Analysis.

Twin/Zygosity Relations (optional)

Upload a twin zygosity sidecar file (CSV or Excel) with columns id1, id2, code (MZ twin, DZ twin, or UZ twin) to mark twin pairs with a connector on the Diagram tab. Leave empty for no twin connectors.

Browse...

No file selected

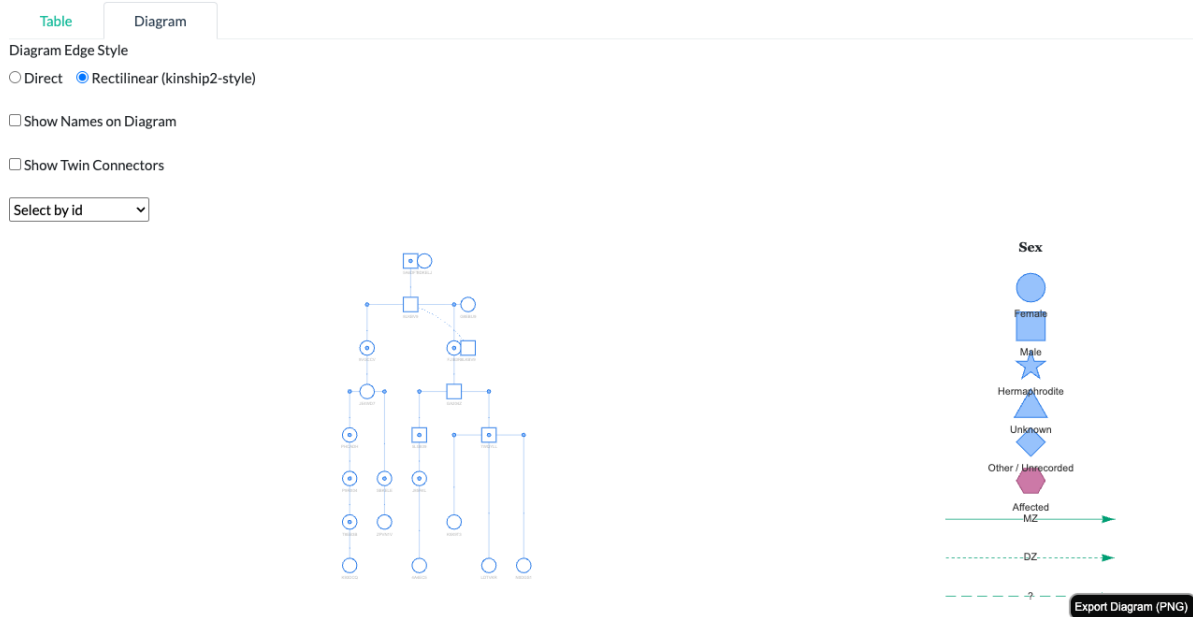


Figure 2: The same 6-animal pedigree with Diagram Edge Style set to Rectilinear – the same relationships, routed in strict right angles.

Both styles show the exact same animals and relationships – the difference is purely routing.

Rectilinear achieves its right angles with extra, invisible “waypoint” nodes and edges (zero size, transparent), which is why its own display cap is lower (400 vs. 750 animals) for the same visual complexity.

Consanguineous mating marker

When a mating pairs two blood-related animals (their kinship coefficient is greater than zero), the two connector lines joining that pair to their shared mating point render thicker and in a distinct, colorblind-safe vermillion (**#D55E00**) – the doubled/thickened mate-line convention traditional pedigree charts use to flag a consanguineous mating at a glance. This marker needs no optional column and no toggle – it is detected directly from the pedigree’s own sire/dam data, via the same kinship computation the rest of the package uses (including any uploaded Twin/Zygosity Relations file described below, for correctness parity). It applies under both edge styles above.

The marker is a real, if visually subtle, cue: the two marked segments sit immediately between each parent’s own icon and the small mating dot next to it, so in a dense, busy diagram it can be easy to miss at a glance even though it renders correctly – the bundled 375-animal example fixture used elsewhere in this article has 28 such consanguineous unions, each contributing its own pair of marked segments. Looking for a specific known relationship, or zooming into one region of a large diagram, makes it much easier to spot than scanning the whole population at once.

Affected-status shading

If the pedigree data includes an optional **affected** column, an individual marked affected renders filled with a distinct color (**#CC79A7**), with a matching “Affected” entry in the legend; individuals marked unaffected, or with unknown/missing affected status, render open/unfilled (white) – matching the standard pedigree-drawing convention that only a filled node means “affected.” Pedigrees without an **affected** column render every node unshaded, as before.

Pedigree Browser

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Focal Animals

IDs of selected focal animals may be manually entered here if analysis of all individuals is not needed. IDs may be pasted from Excel or you can browse for and select a file containing the list of focal animals.

677E7M, BRI2MW, MND3AC

Choose CSV file with focal animals

Browse...

No file selected

Update Focal Animals

☐ Clear Focal Animals

The search field below will search all columns for matches to any text or number entered.

Display Options

☒ Display Unknown IDs

Unknown IDs, by default beginning with a capital U (the format is configurable via `setAutoldFormat()`), are created by the application for all animals with only one parent.

☒ Trim pedigree based on focal animals

Trim the pedigree to include only the ancestors and descendants of the focal animals provided.

Export Pedigree (CSV)

A population must be defined before proceeding to the Genetic Value Analysis.

Twin/Zygosity Relations (optional)

Upload a twin zygosity sidecar file (CSV or Excel) with columns id1, id2, code (MZ twin, DZ twin, or UZ twin) to mark twin pairs with a connector on the Diagram tab. Leave empty

Browse...

No file selected

QC passed! 375 records processed.

Updated focal animals: 3 IDs

Table

Diagram

Diagram Edge Style

☐ Direct ☒ Rectilinear (kinship2-style)

☐ Show Names on Diagram

☐ Show Twin Connectors

Select by id

Sex

Female

Male

Hermaphrodite

Unknown

Other / Unrecorded

Affected MZ

DZ

Export Diagram (PNG)

Figure 3: Three animals from a pedigree that includes an **affected** column: filled (affected), open (unaffected), and open (affected status unknown/NA) – all three unaffected/unknown individuals render identically, open.

Showing names

If the pedigree data includes an optional **name** column, a **Show Names on Diagram** toggle above the diagram (off by default) switches each node's label from id-only to id plus name on a second line. A name longer than 15 characters is truncated with an ellipsis on the diagram itself, with the full name always available in the hover tooltip (see “Interacting with the diagram” below). Not every animal needs a name – one with no name, or a pedigree with no **name** column at all, always renders with just its id, and the **Select by id** search dropdown below always lists ids, never names, regardless of the toggle.

Pedigree Browser

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Focal Animals

IDs of selected focal animals may be manually entered here if analysis of all individuals is not needed. IDs may be pasted from Excel or you can browse for and select a file containing the list of focal animals.

BRI2MW, E80KU8, 677E7M, 6VUC6R

Choose CSV file with focal animals

No file selected

☐ Clear Focal Animals

The search field below will search all columns for matches to any text or number entered.

Display Options

☒ Display Unknown IDs

Unknown IDs, by default beginning with a capital U (the format is configurable via setAutoldFormat()), are created by the application for all animals with only one parent.

☒ Trim pedigree based on focal animals

Trim the pedigree to include only the ancestors and descendants of the focal animals provided.

A population must be defined before proceeding to the Genetic Value Analysis.

Twin/Zygosity Relations (optional)

Upload a twin zygosity sidecar file (CSV or Excel) with columns id1, id2, code (MZ twin, DZ twin, or UZ twin) to mark twin pairs with a connector on the Diagram tab. Leave empty if no twins are expected.

No file selected

QC passed! 375 records processed.
 Updated focal animals: 4 IDs

Table

Diagram

Diagram Edge Style

☐ Direct
 ☒ Rectilinear (kinship2-style)

☒ Show Names on Diagram

☐ Show Twin Connectors

Select by id ▼

Sex

Female

Male

Hermaphrodite

Unknown

Other / Unrecorded

Affected MZ

DZ

-2-

Export Diagram (PNG)

Figure 4: The same small lineage with Show Names on Diagram enabled – most nodes carry an id-plus-name label; the longest is truncated with an ellipsis on the diagram itself (the full name is in its hover tooltip).

Twin/zygosity relations

If a colony records twin births, an optional **Twin/Zygosity Relations** file can be uploaded from the panel to the right of the focal-animal controls – a CSV or Excel file with `id1`, `id2`, and `code` columns (`code` one of "MZ twin", "DZ twin", or "UZ twin"), following `kinship2`'s own twin-code convention. A malformed or inconsistent file (an `id` not in the pedigree, or a declared MZ/DZ pair that does not already share both sire and dam) is rejected with an on-screen notification rather than breaking the diagram; the pedigree renders exactly as it would with no twin data at all until a valid file is supplied.

Once uploaded, a **Show Twin Connectors** toggle above the diagram (off by default) draws a distinctly-styled connector line directly between each declared pair's own diagram nodes, in a colorblind-safe bluish-green (#009E73): solid for a monozygotic (MZ) pair, short-dashed for a dizygotic (DZ) pair, and long-dashed with a “?” label for a pair of unknown zygosity (UZ) – a callback to `kinship2`'s own “?” glyph – with a matching legend entry so the styling is discoverable without hovering over a connector.

Pedigree Browser

- The processed pedigree file shown here will minimally include:
 - Ego ID, Sire ID, Dam ID, Sex, Generation, and Population columns
 - Other potential columns that may appear depending on information provided in the input file include: Birth Date, Exit Date, Age, Group, and Import columns may be present depending on the input file.

Focal Animals

IDs of selected focal animals may be manually entered here if analysis of all individuals is not needed. IDs may be pasted from Excel or you can browse for and select a file containing the list of focal animals.

E06FRB, HV7LZ3, 8GSXTQ, P844CW, BRI2MW, 677E7M

Choose CSV file with focal animals

Browse... No file selected

Update Focal Animals

☐ Clear Focal Animals

The search field below will search all columns for matches to any text or number entered.

Display Options

☒ Display Unknown IDs

Unknown IDs, by default beginning with a capital U (the format is configurable via `setAutoldFormat()`), are created by the application for all animals with only one parent.

☒ Trim pedigree based on focal animals

Trim the pedigree to include only the ancestors and descendants of the focal animals provided.

Export Pedigree (CSV)

A population must be defined before proceeding to the Genetic Value Analysis.

Twin/Zygosity Relations (optional)

Upload a twin zygosity sidcar file (CSV or Excel) with columns id1, id2, code (MZ twin, DZ twin, or UZ twin) to mark twin pairs with a connector on the Diagram tab. Leave empty.

Browse...

QC passed! 375 records processed.

Updated focal animals: 6 IDs

Table Diagram

Diagram Edge Style

☐ Direct ☒ Rectilinear (kinship2-style)

☐ Show Names on Diagram

☒ Show Twin Connectors

Select by id

Sex

Female

Male

Hermaphrodite

Unknown

Other / Unrecorded

Affected MZ

DZ

Export Diagram (PNG)

Figure 5: The same population's diagram with a twin/zygosity file uploaded and Show Twin Connectors enabled – a bluish-green dashed connector runs directly between one declared dizygotic (DZ) pair's own nodes.

Uploading this file does more than draw connectors. A declared monozygotic (MZ) pair’s kinship is corrected to genetic identity throughout the application, not just on this diagram. Once uploaded, the correction is reflected in **Summary Statistics**, **Breeding Group Formation**, and **Genetic Value Analysis** – including for every relative reached through either twin, not just the pair itself – no matter which of those tabs is visited first or whether the **Show Twin Connectors** toggle above is ever switched on (that toggle controls only this diagram’s own rendering). DZ and UZ pairs are unaffected by this correction; only a declared MZ pair’s kinship changes.

Interacting with the diagram

- **Hover** any node to see its ID, sex, generation, sire, dam, and (when present) affected status, without leaving the diagram.
- **Click** a node to re-center the population on that animal, the same as typing its ID into the focal-animals text area – a quick way to explore a different branch of the pedigree. Clicking a duplicate occurrence (see “Node shapes and the legend” above) resolves to the same animal as clicking its main occurrence.
- **Select by id** – the dropdown above the diagram – jumps straight to one animal by ID, dimming every other node except it and its direct connections, useful for finding one animal in a large, busy diagram.
- **Export Diagram (PNG)** – the button in the diagram’s own corner – saves the current view as an image file, useful for husbandry reports, IACUC documents, or presentations.

Script-callable equivalent

Everything this tab renders is built on two exported, script-callable functions that work identically outside the Shiny application: `makePedigreeMatingLayout()` prepares a pedigree’s node/edge data in the mating-unit convention this whole article describes (including the `edgeStyle`, `twinRelations`, and consanguineous-marker behavior above), and `visNetwork::visNetwork()` renders it. A simpler function, `makePedigreeDiagramData()`, remains available for a plain one-node-per-animal diagram without the mating-unit convention – it is no longer what the Shiny app itself uses.

See the “Pedigree Diagram” section of the **Interactive Use of nprcgenomekeeper** vignette for a full, runnable walkthrough – both edge styles, the node/edge data structure, and the twin-relations argument, all reproducing this tab’s rendering exactly (including a genuinely working Export Diagram (PNG) button).

See also

- The **Colony Manager's Guide** article – the Diagram tab in the context of the full Pedigree Browser tab (Table view, focal-animal trimming, export) and the rest of the application.
- The **Interactive Use of nprcgenekeeper** vignette – the same diagram, built step by step from R.
- `makePedigreeMatingLayout()` – prepares the node/edge data this tab renders.
- `makePedigreeDiagramData()` – the simpler, one-node-per-animal alternative.
- `checkTwinRelations()` / `readTwinRelations()` – validate and read a Twin/Zygoty Relations file.
- `kinship()` – pairwise kinship coefficients from a pedigree, including the `twinRelations` correction described above.
- `runGeneKeepR()` – the Shiny app that hosts this tab.